

Week 1 — Mechanical Design

Environment Setup | GitHub Repository Guide

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■ Submission Description

For Week 1, I set up three CAD tools — Tinkercad, Autodesk Fusion, and FreeCAD — and completed a beginner 3D modeling exercise in each, including designing a nameplate, a bracket, and a cylindrical shaft. The technologies used include Tinkercad for browser-based modeling, Autodesk Fusion for parametric CAD, and FreeCAD for open-source offline design. The main challenges faced were adapting to different interfaces and understanding parametric sketching and correct file export formats.

■ Technology Used

Tool	Purpose	Cost
Tinkercad	Browser-based 3D modeling	Free
Autodesk Fusion	Parametric CAD + Cloud storage	Free (personal)
FreeCAD	Open-source offline modeling	Free
GitHub	Documentation & submission	Free

■ What I Built

Tinkercad: Designed a personalized 3D nameplate with my name 'Roshan Koreti' on a flat base plate (80mm x 30mm x 5mm). Exported as STL file ready for 3D printing.

Autodesk Fusion: Created a parametric bracket by sketching a rectangle, adding a mounting hole, and extruding into a 3D solid. Saved to cloud for future edits.

FreeCAD: Modeled a basic cylindrical shaft using Part Design workbench. Sketched a 10mm radius circle and padded it to 50mm height.

■ Challenges Faced

- Adapting to three completely different interfaces in a short period of time.
 - Understanding parametric sketching in Fusion — knowing when to finish a sketch before extruding.
 - Selecting the correct workbench in FreeCAD (Part Design) before modeling tools appear.
 - Learning the correct file export formats — STL for printing, STEP for engineering, .f3d for Fusion.
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■ GitHub Repository Files

README.md — Project description and submission explanation

.gitignore — Ignores CAD backup files, OS junk, and large media files

LICENSE — MIT License — allows others to view and use the work freely
screenshots/ — Folder containing screenshots of all 3 completed exercises